

EE5239 Nonlinear Optimization

Lecture 1: Unconstrained Optimization

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Outline

- 1 Prototype of Unconstrained Optimization
- 2 Optimality Conditions

This Lecture

- Definitions
- Necessary first/second order optimality conditions
- Sufficient optimality conditions
- Existence of optimal solutions
- Quadratic minimization
- Convexity

Announcement

- Will assign the first HW today
 - ① Due **Fri Sept. 20th**
 - ② Problem number according to version 2 of the book; scanned version of the homework problems will be provided.

Unconstrained Optimization

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & x \in \mathbb{R}^n\end{array}$$

- **Objective function** $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **continuous** function
- **Optimization variable** $x \in \mathbb{R}^n$

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- **Unconstrained global minimum** $\hat{x}$: $f(x) \geq f(\hat{x})$ for all $x \in \mathbb{R}^n$

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- **Graphically...**

Existence of Optimal Solution

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- Alternatively, if the level set (for some x^0)

$$\{x \mid f(x) \leq f(x^0)\} \quad (0.3)$$

of a continuous function f is compact, then the global min of

$$\min f(x) \quad (0.4)$$

is attained

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- **Idea.** Use Taylor expansion to analyze local behavior around x

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- Make sense? Check the function $f(x) = x^2$, $f(x_1, x_2) = x_1^2 + x_2^2$
- **Remark.** We call the solutions that satisfy $\nabla f(x^*) = 0$ as **stationary solutions**, or **stationary points**

Checkable Conditions for Local Min

- Let's see the formal proof
- First, let us look at the simple one dimensional case (x is scalar)
- The conditions become

$$f'(x) = 0, \quad f''(x) \geq 0 \quad (0.1)$$

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$$0 \stackrel{(i)}{\leq} \lim_{x^r \downarrow x^*} \frac{f(x^r) - f(x^*)}{x^r - x^*} = f'(x^*) = \lim_{x^r \uparrow x^*} \frac{f(x^r) - f(x^*)}{x^r - x^*} \stackrel{(ii)}{\leq} 0$$

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- (ii): $x^r \leq x^*$, $f(x^r) \geq f(x^*)$ when getting sufficiently close
- To show the second order condition, note

$$0 \stackrel{(iii)}{\leq} \lim_{x^r \rightarrow x^*} \frac{f(x^r) - f(x^*) - f'(x^*)(x^r - x^*)}{(x^r - x^*)^2} = \frac{1}{2} f''(x^*)$$

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- (iii): $f'(x^*) = 0$, and $f(x^r) \geq f(x^*)$ when $x^r \rightarrow x^*$

Checkable Conditions for Local Min

- For higher dimensions, derivation is similar (Prop 1.1.1)
- Consider the one dimensional function $g(\alpha) = f(\mathbf{x}^* + \alpha \mathbf{d})$, where $\mathbf{d} \in \mathbb{R}^n$ is a fixed direction.
- If $\mathbf{x}^*$ is a local min, then $\alpha = 0$ is a local min for $g(\alpha)$ (why?)
- Apply the previous theorem, we have

$$g'(0) = \langle \nabla f(\mathbf{x}^*), \mathbf{d} \rangle = 0, \quad g''(0) = \langle \mathbf{d}, \nabla^2 f(\mathbf{x}^*) \mathbf{d} \rangle \geq 0 \quad (0.2)$$

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- Note $\mathbf{d}$ is an arbitrary direction, then

$$\nabla f(x^*) = 0, \quad \nabla^2 f(\mathbf{x}^*) \succeq 0 \quad (0.3)$$

- For detailed proof, please read Section 1.1 of the text book

Checkable Conditions for Local Min

- **Exercise**: Consider the following functions

$$f(x) = |x|^3, x^3, -|x|^3, x^2, -x^2$$

Check the necessary condition at $x = 0$; plot f

- What have we done so far?
- For a given local min x^* , it **must satisfy** a number of conditions

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- What have we done so far?
- For a given local min x^* , it **must satisfy** a number of conditions
- **Question:** Are these conditions strong enough to “guarantee” local optimality?
- Can we have some simple conditions, so that once they are satisfied by an x^* , then it is local optimum?
- Is the necessary condition also sufficient?

Checkable Conditions for Local Min

- We have the following **sufficient conditions** [makes sense?]

$$\begin{aligned}\nabla f(x^*) &= 0, && \text{(first-order condition),} \\ \nabla^2 f(x^*) &\succ 0, && \text{(second-order condition).}\end{aligned}$$

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- Together they are “sufficient” for local min
- Why? Mean value theorem!
- Let's look at the simple **scalar case**
- The condition becomes $f'(x^*) = 0, f''(x^*) > 0$
- Mean Value Theorem (HW1)?

$$f(x) - f(x^*) = f'(x^{\text{mid}})(x - x^*),$$

$$f(x) - f(x^*) = f'(x^*)(x - x^*) + \frac{1}{2}(x - x^*)^2 f''(x^{\text{mid}}),$$

for some x^{mid} lies in between x, x^*

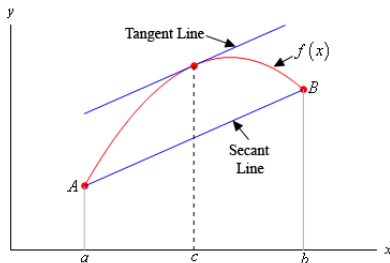
The Mean Value Theorem

- Mean value theorem (MVT)?

$$f(a) - f(b) = f'(c)(a - b),$$

$$f(a) - f(b) = f'(b)(a - b) + \frac{1}{2}(a - b)^2 f''(c),$$

for some c lies in between a, b



Checkable Conditions for Local Min

- Let's look at the scalar case
- MVT suggests there exists an $\alpha \in (0, 1)$

$$f(x) - f(x^*) = f'(x^*)(x - x^*) + \frac{1}{2}(x - x^*)^2 f''(x^* + \alpha(x - x^*)),$$

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- Apply the sufficient condition, for any x sufficiently close to x^*

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- That's why for the sufficient condition strict inequality is needed

Checkable Conditions for Local Min

- Proof for vector version of the sufficient condition (Prop. 1.1.3)
- Vector version of the Mean Value Theorem has to be used
- There exists an $\alpha \in (0, 1)$

$$f(x) - f(x^*) = \langle \nabla f(x^*), x - x^* \rangle + \frac{1}{2} \langle x - x^*, \nabla^2 f(x^* + \alpha(x - x^*)) (x - x^*) \rangle$$

- Apply the sufficient condition, for any x sufficiently close to x^*

$$f(x) - f(x^*) = \frac{1}{2} \langle x - x^*, \nabla^2 f(x^* + \alpha(x - x^*)) (x - x^*) \rangle \succeq 0$$

Intuition: $\nabla^2 f(x^*) \succ 0$, then must exist an neighborhood x close to x^* such that $\nabla^2 f(x) \succeq 0$

Why Optimality Conditions?

- Optimality conditions are useful because:
 - ① provide guarantees for a candidate solution to be optimal (sufficient condition)
 - ② indicate when a point is **NOT** optimal (necessary condition)
- **Guide the design of algorithm**
 - ① Algorithms should look for points achieving the optimality conditions
 - ② Algorithm should stop when the optimality condition is **approximately** satisfied

Use of Optimality Condition

- Let's try to solve the simple problem $\min \frac{1}{2}(x - b)^2$
- Clearly the solution is $x = b$; Is this the same solution predicted by the optimality condition?
- How about $\min -\frac{1}{2}(x - b)^2$? Is the first-order optimality condition sufficient?
- How about the following problems?

$$\min \frac{1}{2}(\mathbf{a}^T \mathbf{x} - b)^2, \quad \min \frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$$

Unconstrained Quadratic Optimization

$$\begin{array}{ll} \text{minimize} & f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q}\mathbf{x} + \mathbf{b}^T \mathbf{x} \\ \text{subject to} & \mathbf{x} \in \mathbb{R}^n \end{array}$$

- Necessary condition for optimality

$$\nabla f(\mathbf{x}) = \mathbf{Q}\mathbf{x} + \mathbf{b} = 0, \quad \nabla^2 f(\mathbf{x}) = \mathbf{Q} \succeq 0$$

Linear Least Squares

$$\begin{array}{ll} \text{minimize} & f(\mathbf{x}) = \frac{1}{2} \|\mathbf{Ax} - \mathbf{c}\|^2 \\ \text{subject to} & \mathbf{x} \in \mathbb{R}^d \end{array}$$

- Recall that $\mathbf{A} \in \mathbb{R}^{n \times d}$, $\mathbf{x} \in \mathbb{R}^d$
- n number of data points, d the dimension of the feature space
- $\mathbf{A}$ may be fat (under-determined), tall (over-determined), or rank-deficient
- Note that comparing with the previous case, $\mathbf{Q} = \mathbf{A}^T \mathbf{A}$, $\mathbf{b} = \mathbf{A}^T \mathbf{c}$
- Necessary condition becomes

$$\mathbf{A}^T \mathbf{Ax}^* - \mathbf{A}^T \mathbf{c} = 0$$

When local-min becomes global-min?

- When local min and global mins are equivalent?
 - ① When the problem belongs to the class of “unconstrained quadratic programs”
 - ② When the problem belongs to a special class called “convex programs”/“convex problems”

Unconstrained Quadratic Optimization

$$\begin{array}{ll}\text{minimize} & f(\mathbf{y}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q}\mathbf{x} + \mathbf{b}^T \mathbf{x} \\ \text{subject to} & \mathbf{x} \in \mathbb{R}^n\end{array}$$

- Necessary condition for (local) optimality

$$\mathbf{Q}\mathbf{x} + \mathbf{b} = 0, \quad \mathbf{Q} \succeq 0 \quad (0.4)$$

- What if the linear system $\mathbf{Q}\mathbf{x} + \mathbf{b}$ is infeasible? What if $\mathbf{Q}$ is not semidefinite?
- Sufficient condition requires $\mathbf{Q} \succ 0$
- **Claim:** The necessary condition is also sufficient; any local optimal solution must be a global optimal solution
- Why? Intuitively, the function cannot have too many curves; More rigorously, next few slides...

Convex function

- A continuous function $f : \mathbb{R}^n \mapsto \mathbb{R}$ is called *convex* if for all $x, y \in \mathbb{R}^n$ and for all $\lambda \in [0, 1]$, we have

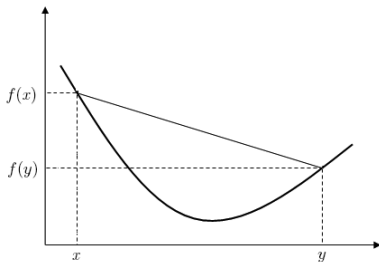
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- Intuitively, a convex function “holds water”



Concave function

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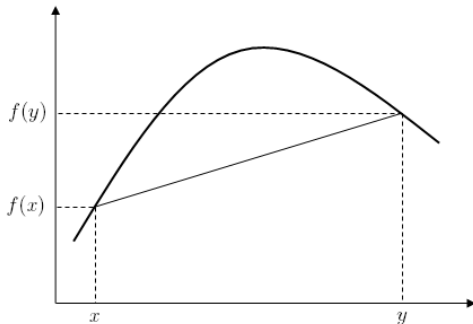
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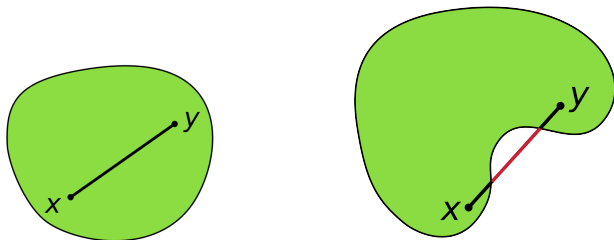
- Question:** A linear function is convex or concave function?



Convex set (need to talk about constraints)

- A set $S \subseteq \mathbb{R}^n$ is **convex** if for any $x, y \in S$ and any $\lambda \in [0, 1]$, we have

$$\lambda x + (1 - \lambda)y \in S$$



- There are **convex sets** and **non-convex sets**
- There is no such thing as a “concave set”
- Question** Is the feasible set generated by **linear constraints** a convex set?

Properties

- If $f(x)$ is a convex function, then $-f(x)$ is a concave function
- If $f_1(x), f_2(x)$ are both convex functions, then $g(x) = f_1(x) + f_2(x)$ are convex as well (prove?)
- If $f_1(x), f_2(x)$ are both convex functions, and $a \geq 0, b \geq 0$, then

$$g(x) = a \times f_1(x) + b \times f_2(x)$$

are convex as well

Properties (Continue)

- Can we use other alternative, and perhaps simpler, ways to characterize the convexity/concavity?
- Yes we can!
- **Given A smooth function with scalar variable; It is convex (resp. concave) if and only if its second-order derivative is nonnegative ≥ 0 (resp. nonpositive ≤ 0)** [show this in HW 1.]
- For vector problems, the above condition becomes its Hessian matrix is positive semidefinite $\nabla^2 f(\mathbf{x}) \succeq 0, \forall \mathbf{x}$
- Go back to the quadratic problems?

Properties (Continue)

- **Second order derivative**, $f''(x)$, shows **how the rate of change of the function is itself changing**

$$f''(x) = \lim_{z \rightarrow x} \frac{f'(z) - f'(x)}{z - x}$$

- **Example** The second derivative of the position of a vehicle with respect to time is the **acceleration**
- **Example** For a function, if it has positive second derivative, then it **curves upwards**
- **Exercise** What are the second order derivatives of the following functions? Can you tell me whether they are convex or concave?

$$y = 5x, \quad y = x^2, \quad y = \log(1 + x), \quad y = x^3$$

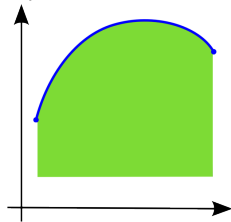
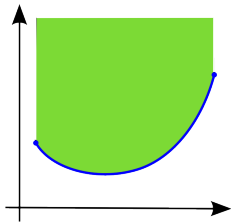
Connection between convex sets and functions

- A function is convex if and only if its **epigraph** (the region above it) is a convex set (prove?)

Connection between convex sets and functions

- A function is convex if and only if its **epigraph** (the region above it) is a convex set (prove?)
- function is **concave** if and only if its **hypograph** (the region below it)

it) is a convex set.



Convex Functions

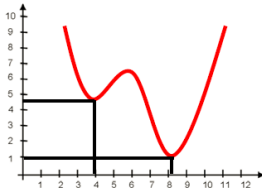
- Please determine the convexity of the following functions

$$x; \quad x^2; \quad x^3; \quad x^4; \quad \frac{1}{x}; \quad \log(x); \quad \sin(x); \quad \frac{\sin(x)}{x}, \quad x \in (0, 1).$$

Why studying all this?

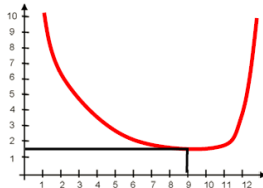
- Why we studying all this?
- Minimizing a convex function is always easy
- Necessary condition becomes sufficient

Non convex function



Local minimum might not
be a global minimum

Convex function



Local minimum is a global
minimum

Properties

- Generally speaking, for the following types of **unconstrained problems**

$$\min f(x), \quad \max f(x) \quad (0.5)$$

we have the following understanding:

	convex function	concave function
max	hard	easy
min	easy	hard

- Why? Simple intuitive argument using Taylor expansion?

Properties

- Let's pick one problem and see why

Properties

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- Consider minimizing a convex function

Properties

- Let's pick one problem and see why
- Consider minimizing a convex function
- **Claim** any local minimum is global minimum

Properties

- Let's pick one problem and see why
- Consider **minimizing a convex function**
- **Claim** any local minimum is global minimum
- Proven using definition of convexity
 - Suppose $\bar{x}$ is local but not global minimum
 - Then there exists $f(x) < f(\bar{x})$
 - Due to convexity, for any $c \in (0, 1)$

$$\begin{aligned}f[c\bar{x} + (1 - c)x] &\leq cf(\bar{x}) + (1 - c)f(x) \\ &< cf(\bar{x}) + (1 - c)f(\bar{x}) \\ &= f(\bar{x})\end{aligned}$$

Properties

- Let's pick one problem and see why
- Consider **minimizing a convex function**
- **Claim** any local minimum is global minimum
- Proven using definition of convexity
 - Suppose $\bar{x}$ is local but not global minimum
 - Then there exists $f(x) < f(\bar{x})$
 - Due to convexity, for any $c \in (0, 1)$

$$\begin{aligned}f[c\bar{x} + (1 - c)x] &\leq cf(\bar{x}) + (1 - c)f(x) \\&< cf(\bar{x}) + (1 - c)f(\bar{x}) \\&= f(\bar{x})\end{aligned}$$

- Contradiction to $\bar{x}$ being local optimal (why?)